

Summary

Distributed systems research engineer focused on Byzantine fault tolerance and consensus, with production leadership on large-scale blockchain infrastructure. Technical lead for consensus and execution on Aptos, a high-performance public blockchain; designed BFT protocols including Raptr and Shoal++ (NSDI 2025).

Experience

Aptos Labs

Lead Research Engineer

Jul. 2022 – Present

- Technical lead for consensus and execution on the Aptos blockchain; owned architecture and roadmaps for BFT protocols, performance, reliability, and node operations.
- Led mainnet performance improvements to 30k+ TPS, sub-500 ms p99 latency, and $\approx 70\%$ block time reduction to sub-30 ms.
- Designed and developed Raptr, Shoal++, and Shoal BFT protocols (Shoal++ at NSDI 2025). Raptr reached 250k+ benchmark TPS at 750 ms latency in a 100-node geo-distributed deployment.
- Landed 2 adopted AIPs; shipped Encrypted Mempool and Distributed Key Generation; audited and resolved multiple \$100k+ bug bounties pre-exploit.
- ~ 200 PRs reviewed annually; led design and code review standards; conducted interviews, led debrief panels, and mentored interviewers.

LinkedIn Corporation

Site Reliability Engineer

Apr. 2021 – Jul. 2022

- Built analytics ingestion and stream-processing pipelines for the Product Experience Monitoring platform, collecting experience telemetry from 800+ million members daily across various LinkedIn products.
- Optimized the distributed stream-processing system to handle millions of events per second, enabling real-time member experience insights for product and reliability teams.

Virginia Tech

Graduate Research Assistant

Sep. 2015 – Apr. 2021

- Research in the System Software Research Group on scalable Byzantine state machine replication and low-latency geo-distributed consensus (Ph.D. dissertation and M.S. thesis).
- Designed globally scalable Byzantine and hybrid fault-tolerant consensus protocols; developed dual fault-tolerant protocols with resilience–efficiency tradeoffs.

Virginia Tech

Graduate Teaching Assistant

- Teaching Assistant for Multiprocessor Programming (Graduate-level) course during the Fall 2016, 2017 and 2019 semesters.
- Teaching Assistant for Applied Software Design (Junior-level) course during the Spring 2016 semester.
- Designed course assignments and projects, evaluated student submissions, conducted hands-on lab classes, and held weekly office hours.

LinkedIn Corporation

Site Reliability Engineering Intern

May. 2018 – Aug. 2018

- Developed the minimal viable version of Scattershot, a distributed log indexing and searching system with live-streaming support.
- Designed and implemented a high-performance dashboard for live-streaming thousands of log events per second directly from production hosts.
- Marketed the product to various internal teams and on-boarded the early adopters of the product.

Facebook Inc.

Production Engineering Intern

May. 2017 – Aug. 2017

- Designed and implemented a regression testing framework for Warm Storage, Facebook's data warehouse storage system.
- Identified and fixed scalability issues in multiple critical micro-services that impacted production using the framework.
- Integrated the testing suite as part of the continuous integration pipeline.

Education

Virginia Polytechnic Institute and State University (Virginia Tech) 2015 – 2021

Ph.D. in Computer Engineering

Dissertation: Scalable Byzantine State Machine Replication

Virginia Polytechnic Institute and State University (Virginia Tech) 2015 – 2017

M.S. in Computer Engineering

Thesis: A Low-latency Consensus Algorithm for Geographically Distributed Systems

Virginia Polytechnic Institute and State University (Virginia Tech) 2011 – 2015

B.S. in Computer Engineering, *Summa Cum Laude*

Publications

- [1] **B. Arun**, Z. Li, F. Suri-Payer, S. Das, and A. Spiegelman. Shoal++: High Throughput DAG BFT Can Be Fast and Robust! In *22nd USENIX Symposium on Networked Systems Design and Implementation (NSDI 25)*, pages 813–826, Philadelphia, PA, April 2025. USENIX Association.
- [2] A. Spiegelman, **B. Arun**, R. Gelashvili, and Z. Li. Shoal: Improving DAG-BFT Latency And Robustness. In *Financial Cryptography and Data Security*. Springer International Publishing, 2024. <https://arxiv.org/abs/2306.03058>.
- [3] A. Tonkikh, **B. Arun**, Z. Xiang, Z. Li, and A. Spiegelman. Raptr: Prefix Consensus for Robust High-Performance BFT. *CoRR*, abs/2504.18649, 2025. <https://arxiv.org/abs/2504.18649>.

- [4] Z. Xiang, Z. Li, **B. Arun**, T. Zhang, and A. Spiegelman. Zaptos: Towards Optimal Blockchain Latency. *CoRR*, abs/2501.10612, 2025. <https://arxiv.org/abs/2501.10612>.
- [5] **B. Arun** and B. Ravindran. Scalable Byzantine Fault Tolerance via Partial Decentralization. *Proc. VLDB Endow.*, 15(9):1739–1752, May 2022.
- [6] **B. Arun** and B. Ravindran. DuoBFT: Resilience vs. Efficiency Trade-off in Byzantine Fault Tolerance. *CoRR*, abs/2010.01387, 2020. <https://arxiv.org/abs/2010.01387>.
- [7] **B. Arun**, S. Peluso, R. Palmieri, G. Losa, and B. Ravindran. Taming the Contention in Consensus-based Distributed Systems. *IEEE Transactions on Dependable and Secure Computing*, 18(6):2907–2925, 2021.
- [8] M. Garg, S. Peluso, **B. Arun**, and B. Ravindran. Generalized Consensus for Practical Fault Tolerance. In *Proceedings of the 20th International Middleware Conference, Middleware '19*, page 55–67, New York, NY, USA, 2019. Association for Computing Machinery.
- [9] **B. Arun**, S. Peluso, and B. Ravindran. ezBFT: Decentralizing Byzantine Fault-Tolerant State Machine Replication. In *2019 IEEE 39th International Conference on Distributed Computing Systems (ICDCS)*, pages 565–577, July 2019.
- [10] **B. Arun**, S. Peluso, R. Palmieri, G. Losa, and B. Ravindran. Speeding up consensus by chasing fast decisions. In *2017 47th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, pages 49–60, 2017.
- [11] **B. Arun**, S. Hirve, R. Palmieri, S. Peluso, and B. Ravindran. Speculative Client Execution in Deferred Update Replication. In *Proceedings of the 9th Workshop on Middleware for Next Generation Internet Computing, MW4NG '14*, New York, NY, USA, 2014. Association for Computing Machinery.

Talks

- Shoal++: High throughput DAG BFT can be fast and robust! NSDI 2025
- Scalable Byzantine Fault Tolerance via Partial Decentralization VLDB 2022
- Speeding up Consensus by Chasing Fast Decisions Air Force OSR SRD 2017
- ezBFT: Decentralizing Byzantine Fault-Tolerant State Machine Replication ICDCS 2019
- Speeding up Consensus by Chasing Fast Decisions DSN 2017
- Leaderless Consensus Data Skeptic Podcast

Professional Activities

- Reviewer, Transactions on Computers IEEE TC 2024
- Reviewer, IEEE Transactions on Dependable and Secure Computing IEEE TDSC
- Reviewer, Advances in Financial Technologies AFT 2023
- Reviewer, Financial Cryptography and Data Security FC 2023
- Reviewer, Symposium on the Principles of Distributed Computing PODC 2023
- Reviewer, 41st International Symposium on Reliable Distributed Systems SRDS 2022

- Reviewer, ACM Principles and Practice of Parallel Programming *PPoPP 2022*
- Reviewer, Conference on Distributed Computing and Intelligent Technology *ICDCIT 2022*
- Reviewer, Symposium on Stabilization, Safety, and Security of Dis. Systems *SSS 2020*
- Shadow Program Committee, Eurosys *Eurosys 2020*
- Reviewer, IEEE Transactions on Dependable and Secure Computing *TDSC 2020*
- External Reviewer, IEEE Conference on Dependable Systems and Networks *DSN 2020*
- Reviewer, International Conference on Principles of Distributed Systems *OPODIS 2019*
- Reviewer, International Symposium on Distributed Computing *DISC 2019*
- Reviewer, International Symposium on Distributed Computing *DISC 2018*
- Reviewer, International Conference on Principles of Distributed Systems *OPODIS 2017*